

Jerry Cai

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechanical Engineering

Waterloo, ON

2026 – 2031

EXPERIENCE

Mechanical Team Member - Dynamics

2026 – TBD

University of Waterloo Midnight Sun Solar Car Team

Waterloo, ON

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Kumon Tutor

2024 – 2025

Kumon

Calgary, AB

- Demonstrated instructional leadership by simplifying complex mathematical and literacy concepts into digestible steps tailored to diverse learning levels
- Refined time management and organizational skills by balancing the needs of multiple students simultaneously while maintaining a productive and disciplined learning atmosphere
- Gained experience in progress assessment by evaluating student work critically and providing actionable feedback to encourage continuous improvement

Line Cook

2024 – 2025

Calgary Stampede

Calgary, AB

- Assisted in fast-paced professional kitchen environments, working directly under qualified and experienced chefs, gaining hands-on experience in food preparation and interaction with customers
- Worked and created positive relationships with kitchen staff to ensure efficiency and quality food production in a fast-paced and high-volume open restaurant setting, demonstrating hospitality and positivity to customers
- Supported events by setting up kitchen functions, unloading and packing cargo, and maintaining kitchen areas

PROJECTS

Circadian Rhythm Optimizer | Arduino, Fusion 360, Fritzing

2026

- Engineered an automated circadian-rhythm roller blind opener using an Arduino Uno R3, DS3231 RTC module, and photoresistor-based light sensing to trigger blind actuation on time and ambient-light thresholds
- Designed a stepper-motor drive train (28BYJ-48 + ULN2003 driver) with a custom 3D-printed sprocket to convert rotational motion into cord-pulling force, iterating through multiple failed transmission methods
- Diagnosed a torque deficiency (34.3 mN·m actual vs. load requirement) and specified a NEMA-17 stepper (40–60 N·cm) as a corrective fix for a future iteration
- Prototyped circuit schematics in Fritzing and modeled a protective enclosure in Fusion 360, documenting the full design-build-test cycle

Sedentary Squat Counter Reminder | C#, WPF (.NET Framework), Visual Studio, Kinect SDK

2026

- Built a sedentary-activity reminder app in C# / WPF (.NET Framework) using an Xbox 360 Kinect sensor for real-time 3D skeletal tracking and joint-angle detection
- Implemented squat-repetition logic via dot-product knee-angle calculations in 3D Euclidean space, with debounce filtering to prevent false reps from partial movements
- Designed a check-in system that locks the screen on a configurable interval and only releases control once a user-defined rep goal is met, enforcing active breaks
- Debugged a calibration bug causing rep-counter resets, resolving it with conditional reset logic in the counter's state-management class

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, MATLAB, HTML/CSS

CAD & Simulation: AutoCAD, Fusion 360, SolidWorks (CSWA in progress)

Developer Tools: Git/GitHub, Visual Studio, VS Code, Fritzing

Embedded/Hardware: Arduino (Uno R3), Kinect SDK, stepper motor drivers, RTC modules